Normalisation

(every length $\div R$; wall plane at $\bar{w}=0$, contact at $\bar{w}=1$, so $\bar{w}\geq 1$)

$\bar{w}$	probe-centre position
$\bar{w}_s$	contact position, unknown; nominal 1 (surface itself at $\bar{w}_s-1$)
$(\bar{w}-\bar{w}_s)$	gap between the two surfaces; 0 at contact
$\bar{a}=a/a_o$	motion gain, $\in(0,1)$
$b,\ p$	shape parameters; $\theta=(b,\ p,\ \bar{w}_s)$ in both forms

Truth

(Brenner 1961 series, $\bar{a}=1/c_\perp$)

$$\bar{a}(\bar{w})=1-\tfrac{9}{8}\bar{w}^{-1}+\tfrac{1}{2}\bar{w}^{-3}-\tfrac{57}{100}\bar{w}^{-4}+\tfrac{1}{5}\bar{w}^{-5}+\tfrac{7}{200}\bar{w}^{-11}-\tfrac{1}{25}\bar{w}^{-12}$$

$$\text{near wall}: \quad \bar{a}\simeq(\bar{w}-\bar{w}_s), \qquad\qquad \text{far field}: \quad 1-\bar{a}\simeq\frac{9/8}{(\bar{w}-\bar{w}_s)}$$

Form A

(the proposal)

$$\boxed{\bar{a}(\bar{w}-\bar{w}_s)=1-e^{-\frac{b}{p+1}(\bar{w}-\bar{w}_s)^{p+1}}}$$

$$\bar{a}'=b\,(\bar{w}-\bar{w}_s)^{p}\,e^{-\frac{b}{p+1}(\bar{w}-\bar{w}_s)^{p+1}}$$

$$\boxed{\begin{aligned} J_b&=\bar{a}'\left[\frac{1}{b}-\frac{(\bar{w}-\bar{w}_s)^{p+1}}{p+1}\right]\\ J_p&=\bar{a}'\left[\left(1-\frac{b}{p+1}(\bar{w}-\bar{w}_s)^{p+1}\right)\ln(\bar{w}-\bar{w}_s)+\frac{b}{(p+1)^2}(\bar{w}-\bar{w}_s)^{p+1}\right]\\ J_{w_s}&=\bar{a}'\left[b\,(\bar{w}-\bar{w}_s)^{p}-\frac{p}{(\bar{w}-\bar{w}_s)}\right]\end{aligned}}$$

Two limits of Form A

near wall, $\bar{a}\simeq\frac{b}{p+1}(\bar{w}-\bar{w}_s)^{p+1}$ must be linear in $(\bar{w}-\bar{w}_s)$:

$p=0,\qquad b=1$

far field, $1-\bar{a}=e^{-\frac{b}{p+1}(\bar{w}-\bar{w}_s)^{p+1}}$ must decay as a power; for $p+1>0$ it decays faster than every power:

$p=-1$

$$\boxed{p=0\ \wedge\ p=-1\quad:\ \text{no solution, for any }(b,\bar{w}_s)}$$

Form B

$$\boxed{\begin{aligned}\bar a\big(\bar w-\bar w_s,\,b,\,p\big)&=1-\left(1+\frac{\bar w-\bar w_s}{b}\right)^{-p} &&(\bar w_s-1:\ \text{unknown surface position})\\ \bar a\big(\bar w-\bar w_s,\,b_w,\,p_w\big)&=1-\left(1+\frac{\bar w-\bar w_s}{b_w}\right)^{-p_w}\\ \bar a\big(\bar w_d-\hat{\bar w}_s,\,\hat b_w,\,\hat p_w\big)&=1-\left(1+\frac{\bar w_d-\hat{\bar w}_s}{\hat b_w}\right)^{-\hat p_w}\end{aligned}}$$

equivalently, and the writing used for the derivatives below:

$$\bar a(\bar w-\bar w_s)=1-\left(1+\frac{(\bar w-\bar w_s)}{b}\right)^{-p}=1-\left[\frac{b}{(\bar w-\bar w_s)+b}\right]^{p}$$

$$\bar a'=p\,b^p\,\big((\bar w-\bar w_s)+b\big)^{-p-1}$$

$$\boxed{\begin{aligned}J_b&=\bar a'\left[\frac{p}{b}-\frac{p+1}{(\bar w-\bar w_s)+b}\right]\\ J_p&=\bar a'\left[\frac{1}{p}+\ln\frac{b}{(\bar w-\bar w_s)+b}\right]\\ J_{w_s}&=\bar a'\,\frac{p+1}{(\bar w-\bar w_s)+b}\end{aligned}}$$

Two limits of Form B

near wall, $(\bar w-\bar w_s)\ll b$, drop the square and higher terms of $(1+(\bar w-\bar w_s)/b)^{-p}\simeq 1-p\,(\bar w-\bar w_s)/b$, then match $\bar a\simeq(\bar w-\bar w_s)$:

$$\bar a\simeq\frac{p}{b}\,(\bar w-\bar w_s)\qquad\Longrightarrow\qquad\frac{p}{b}=1$$

far field, $(\bar w-\bar w_s)\gg b$, drop the 1 against $(\bar w-\bar w_s)/b$, then match $1-\bar a\simeq\frac{9/8}{(\bar w-\bar w_s)}$:

$$1-\bar a\simeq\left(\frac{(\bar w-\bar w_s)}{b}\right)^{-p}=b^p\,(\bar w-\bar w_s)^{-p}\qquad\Longrightarrow\qquad p=1,\quad b=\tfrac{9}{8}$$

$$\boxed{\text{over-determined by }12.5\%\ (b=1\text{ vs }b=\tfrac{9}{8}),\ \text{carried by one parameter}}$$

Under test

$$\theta[k+1]=\theta[k],\qquad Q=0$$

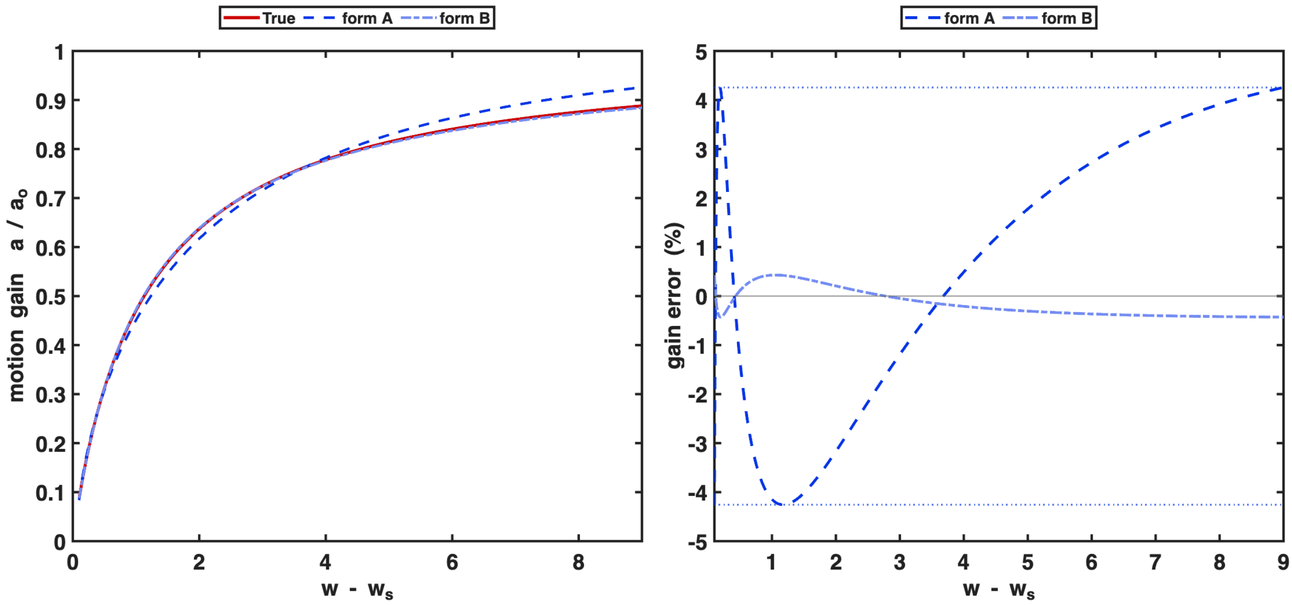
Best-case fit

($\bar{w} \in [1.1, 10]$, i.e. gap $(\bar{w} - \bar{w}_s) \in [0.1, 9]$; minimax, 2×10^4 samples)

$$\sup \left| \frac{\bar{a}_{\text{model}}}{\bar{a}_{\text{true}}} - 1 \right|$$

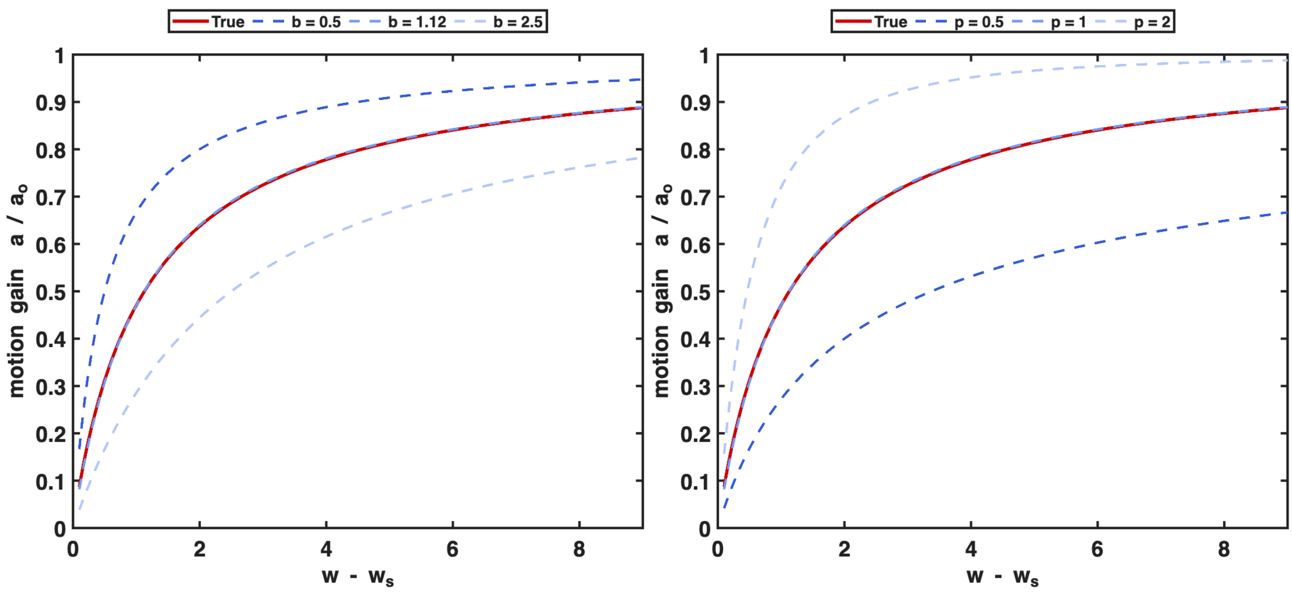
form	parameters	sup [%]	$\bar{w}_s - 1$ [μm]	equiosc. / starts
A	b=0.406, p=$-$0.346, $\bar{w}_s$=1.0502	4.256	+0.113	4/4, 6/6
B	b=1.0566, p=0.9563, $\bar{w}_s$=0.9936	**0.429**	**$-$0.015**	4/4, 5/5

Form B against its derived values: $b = 1.0566$ vs $9/8 = 1.125$, $p = 0.9563$ vs 1.



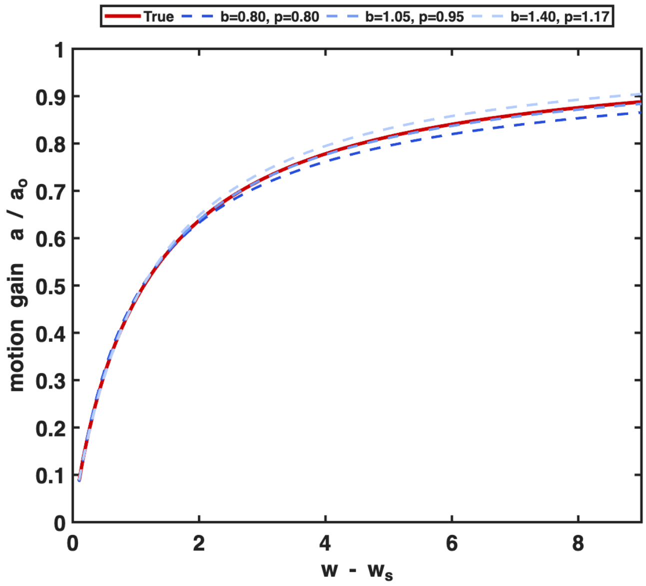
What b and p do

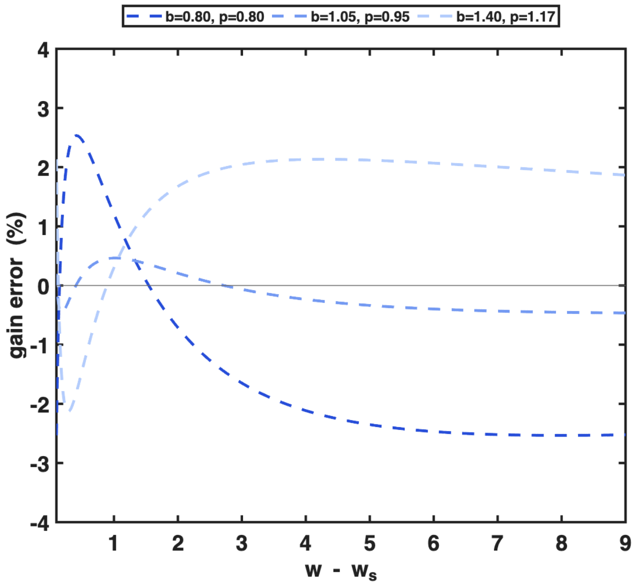
(Form B; left $p = 1$ with b swept, right $b = 9/8$ with p swept; red is the truth, the middle curve is the derived value)



The two are not separable

(three parameter sets from the floor of the error valley, each with its own best $\bar{w}_s$)





b	p	$\bar{w}_s$	p/b	sup [%]
0.80	0.8010	1.0060	1.0013	2.533
1.05	0.9528	0.9941	0.9074	0.464
1.40	1.1704	0.9837	0.8360	2.133

$$b: \ 1.75\times, \qquad p: \ 1.46\times, \qquad \text{gain error}: < 2.6\%, \qquad \text{corr}(\ln b, \ln p) = 0.986$$